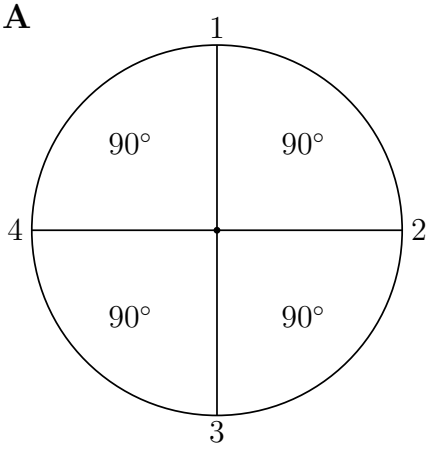
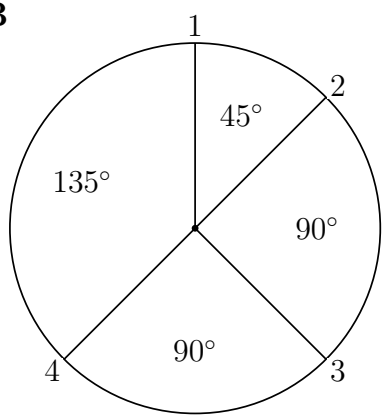
Use the same marked front for all mountain (M) and valley (V) labels. Fold every listed ray fully flat, with no extra creases or cuts. Layers may touch but may not pass through one another.

Problem 1: Decide which patterns can close flat. Record a successful fold with M/V labels for each one you can make; leave any undecided pattern marked with a question mark.

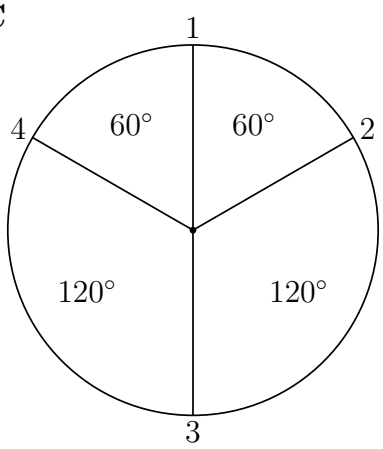
A



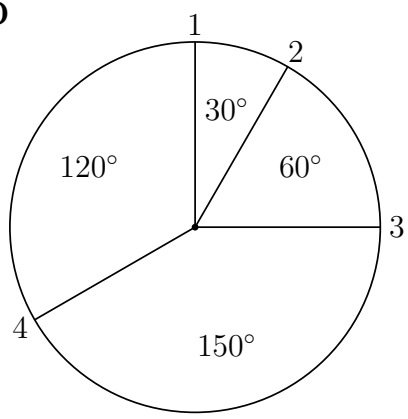
B



C

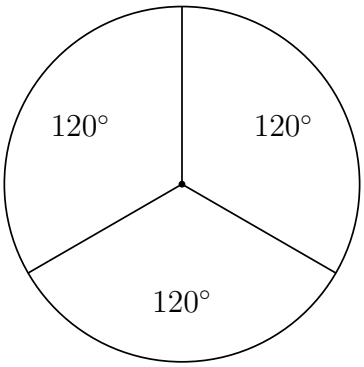


D

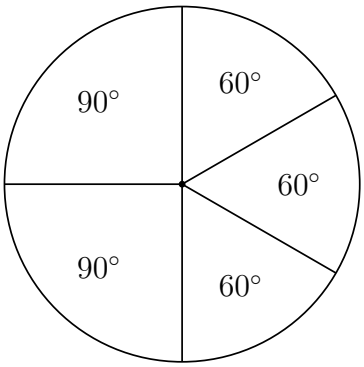


Problem 2: Can any of these patterns close flat along every ray? Explain an answer that covers all three patterns.

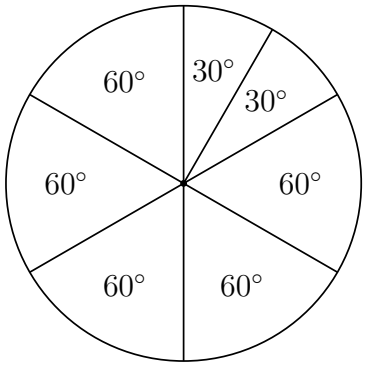
A



B

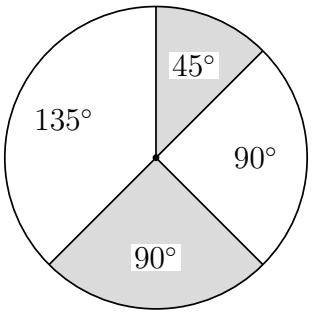


C

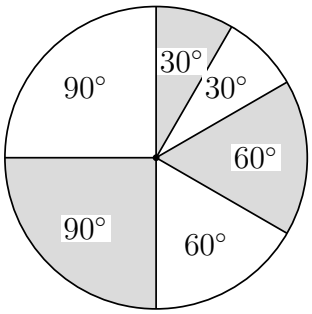


Problem 3: Explain why every other sector must total 180° in any flat-folding pattern with an even number of rays. Which patterns does your explanation rule out?

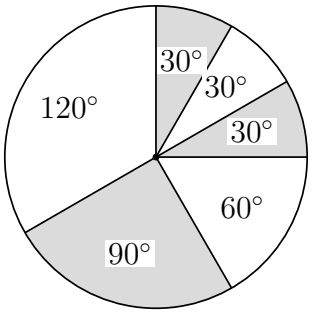
A



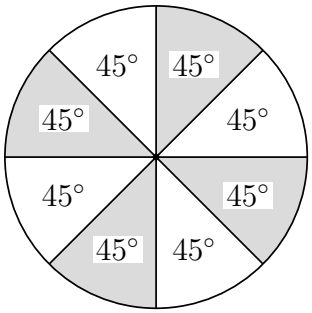
B



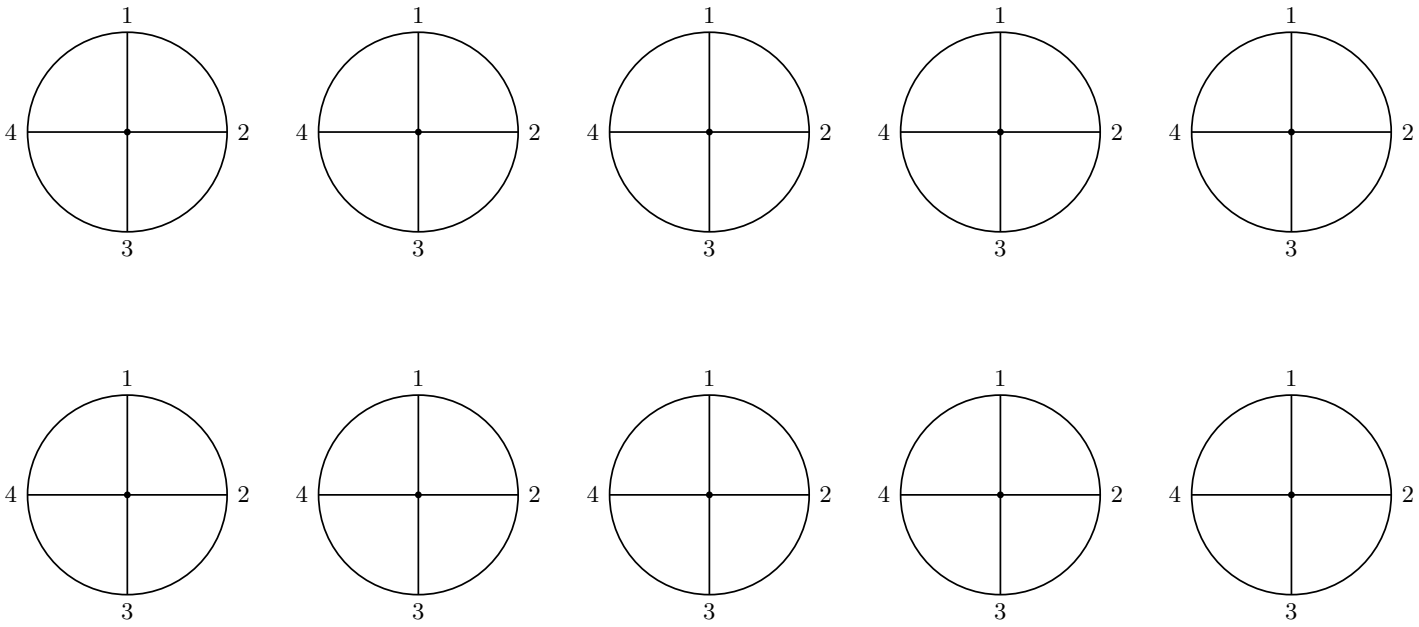
C



D

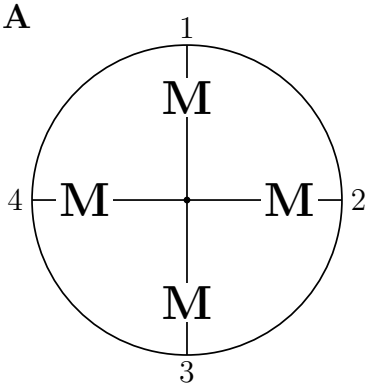


Problem 4: Find as many mountain/valley labelings of the four-right-angle disk that can close flat as you can.

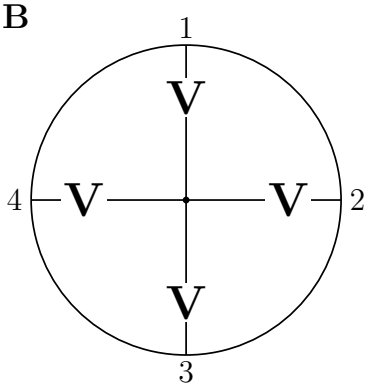


Problem 5: Can any of these labelings close flat? Explain why your folds from Problem 4 and the patterns you rule out account for every M/V labeling of the four-right-angle disk.

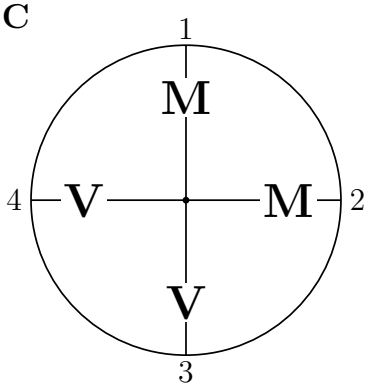
A



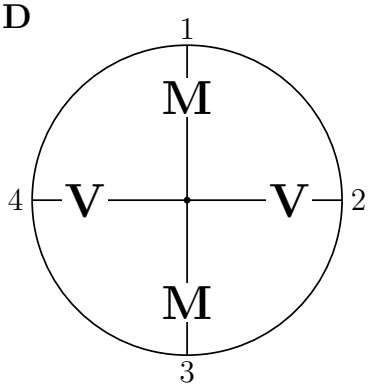
B



C



D



Problem 6: Use the angles $30°, 30°, 60°, 60°, 90°, 90°$ once each, clockwise around one center. How many different orders start with $30°$ and have alternating totals of $180°$? Explain why your count is complete. Swapping equal angles does not make a new order.

